

A low-angle, upward-looking photograph of a tall telecommunications tower. The tower is constructed from a complex lattice of red and white metal beams. Several large, white, parabolic satellite dishes and other antennas are mounted at different heights along the tower's structure. The background is a clear, solid blue sky. A semi-transparent purple and red gradient banner is overlaid at the bottom of the image, containing the title text.

# An overview of The Open RAN system

# Abstract:

Open RAN is the movement in wireless telecommunications to create intelligent, open, virtualized, and fully interoperable Radio Access Networks (RAN) by disaggregating RAN functionality, hardware, and software, creating open interfaces between them, and providing a framework for intelligent control and optimization. This whitepaper provides an overview of the Open RAN system architecture as defined by the O-RAN alliance. Different O-RAN functional splits, benefits, deployments, opportunities, and challenges are discussed in this white paper.

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# Table of Content

|                                          |    |
|------------------------------------------|----|
| What is Open RAN .....                   | 4  |
| The ORAN Ecosystem .....                 | 5  |
| ORAN Architecture .....                  | 8  |
| ORAN Functional Splits .....             | 11 |
| ORAN Benefits .....                      | 14 |
| ORAN Deployments and Opportunities ..... | 14 |
| ORAN Challenges .....                    | 15 |
| Conclusion.....                          | 16 |
| References.....                          | 16 |
| Glossary.....                            | 16 |
| About the Author .....                   | 17 |
| About Sasken .....                       | 17 |



# What is Open RAN?

Open RAN is a Radio Access Network (RAN) with standardized elements and interfaces. O-RAN is the standardization of Open RAN by the ORAN Alliance. Traditionally, RAN is composed of a 'monolithic' architecture, which is deployed, depending on the scenario, either as a single unit (BBU+RF) at the cell site or as a centralized Base Band Unit (BBU) with Remote Radio Heads (RRH/RU). In the latter case, BBU managed all the processing (PHY, MAC, RLC, PDCP, and RRC), and the RRH handled RF processing. Usually, a single vendor was responsible for the RAN architecture (hardware, software, and interfaces). The RAN architecture was fully proprietary, and its interfaces were closed. It offered the least flexibility in terms of scalability and control. Updates such as feature additions and release upgrades involve long cycles and often require HW upgrades. This, in turn, had a significant impact on the TCO.

To increase the deployment flexibility for 5G NR, 3GPP split the functionality of BBU into a Centralized Unit (CU) and a Distributed Unit (DU). O-RAN builds upon these architecture principles and further divides the DU into O-DU and O-RU. Interfaces between O-CU and O-DU (F1) and O-DU and O-RU (7.2x) are fully defined and open, ensuring multi-vendor interoperability. This enables operators to mix and match entities from different vendors as per their requirements. They can at any time change a specific vendor or add new vendors without disrupting the existing network. O-RAN also provides the possibility of intelligent control (programmability and performance optimization at different levels) by incorporating essential elements into its architecture framework (RAN Intelligent Control (RIC)). Going further, O-RAN also aims for disaggregation of hardware and software. Disaggregation refers to the concept of implementing desired

functionality (for example, network functionality) in software with minimal dependence on specific hardware. The commercial off-the-shelf (COTS) hardware should be able to run the software functionality from any vendor. In some cases, specific accelerators may be needed to achieve the required performance (ex., Forward Error Correction (FEC) acceleration in Layer 1). O-RAN provides the required flexibility and abstraction in those cases. Disaggregation of HW-SW also helps in selecting best-of-breed software solutions from different vendors while retaining the investments made in HW or vice versa. It also helps in independently choosing the HW and SW providers purely based on performance and cost, thereby fostering open and healthy competition.

The below diagram shows the O-RAN approach for BBU disaggregation.

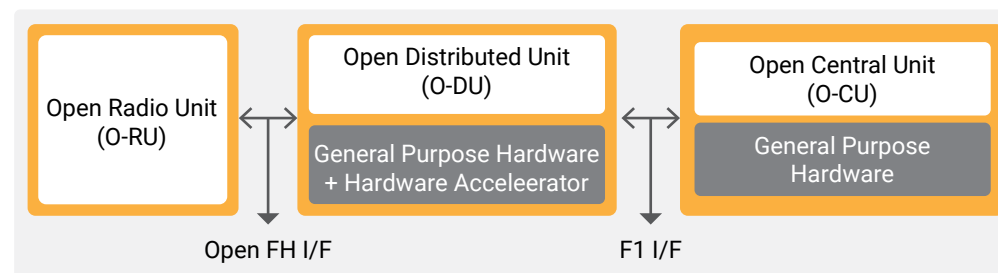


Figure 1: Open RAN Approach for BBU disaggregation.



# The ORAN Ecosystem

The other goal of O-RAN is network automation. This results in a simplification of Operation and Maintenance (O&M). Artificial Intelligence/Machine Learning (AI/ML) find applications in several domains, and telecom is no exception. O-RAN architecture provides a framework (example: Near-RT and non-RT RIC apps) to integrate AI/ML techniques in areas of performance optimization and control. AI/ML/DL (Deep Learning) techniques are also finding their way into realizing network functions at different protocol layers and are set to completely revolutionize the existing SW and HW architectures of RAN systems, and O-RAN is designing its architecture and functionality to embrace this.

There are several industry groups focused on Open RAN. The two main organizations driving the Open RAN movement are:

## **Telecom Infra Project (TIP)**

TIP supports the development of a disaggregated and interoperable 2G/3G/4G/5G-NR Radio Access Network (RAN) solution based on service provider requirements. TIP's activities include defining, building, testing, releasing, and deploying Open RAN solutions. The focus is not on the development of interface standards but rather on referencing those standards (ex. 3GPP, O-RAN) to define requirements, use cases, deployment guidelines, and test outcomes. The aim is to advance Open RAN deployments via trials, plugfests, and live field deployments. TIP community labs, which are located all over the world and sponsored by individual TIP participants, are used for testing, validating, and integrating ORAN solutions to accelerate deployments.



## ORAN-Alliance

ORAN-Alliance was formed in 2018 by China Mobile, AT&T, NTT Docomo, Deutsche Telekom, and Orange and is now a world-wide community of operators, vendors, and academic institutions. The goal of this alliance is twofold. One is to enable small operators to introduce their services or customize the network to suit their needs via cloud-scale economies. The second is to automate the networks, which results in the simplification of O&M. This group develops, drives, and enforces standards (e.g., fronthaul specifications) to ensure that equipment from multiple vendors interoperates with each other. This enables a rich supplier ecosystem with faster innovation cycles to improve the efficiencies of deployment, operations, and maintenance, which in turn results in a reduction of CAPEX and OPEX for the RAN networks. To achieve these objectives, the ORAN Alliance carries out specification, development, testing, and integration activities through their working groups, the software community, and their support to O-RAN members in the testing and integration of their solutions.

The ORAN Alliance has a total of 11 working groups, as detailed in the table below:

| O-RAN Technical Work Group                                                             | High level Objectives of the Work Group                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WG1 - Use Cases and Overall Architecture Workgroup.                                    | Overall architecture specification, including O-RAN N/W slicing architecture, definition of O-RAN functions and corresponding interfaces, and use cases that are enabled by O-RAN                                                                                           |
| WG2 - Non Realtime RAN Intelligent Controller (Non-RT RIC) and A1 Interface Workgroup. | Specification of non-RT RIC and A1 interface, whose objectives are to support non-real time intelligent radio resource management, optimization of procedures and policies, providing AI/ML models to RAN functions, and policy-based guidance of the near-RT RIC functions |
| WG3 - Near-real-time RIC (Near-RT RIC) and E2 Interface Workgroup.                     | Specification of requirements, architecture, use cases, and conformance tests for the near-RT RIC and E2 interfaces; API definition between the near-RT RIC platform and xAPPs                                                                                              |
| WG4 - Open Fronthaul Interfaces Workgroup                                              | Specification of the open FH interface (Control/User/Synchronization/Management Plane (C/U/S/M) protocols) between O-DU/SMO and O-RU Conformance, IOT specifications, and IOT profiles                                                                                      |
| WG5 - Open F1/W1/E1/X2/Xn Interface Workgroup                                          | Ensuring interoperability of the 3GPP-defined interfaces for the higher layer split by performing gap analysis and defining an interoperability test specification. Definition of management models for O&M support within the overall framework of WG10                    |



|                                                  |                                                                                                                                                                                                                                                                    |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WG6 - Cloudification and Orchestration Workgroup | Identify use cases for h/w/s/w decoupling benefits, develop requirements and reference designs for the cloud platform (virtualization infrastructure, container/vm orchestration, acceleration abstraction layers), and develop lifecycle flows for orchestration. |
| WG7 - White-box Hardware Workgroup               | Specification of high performance, spectral and energy efficient base station HW reference designs                                                                                                                                                                 |
| WG8 - Stack Reference Design Workgroup           | O-CU, O-DU Software architecture, functional blocks, and API definitions. Stack interoperability Test Specification.                                                                                                                                               |
| WG9 - Open X-haul Transport Workgroup            | Defines requirements and best practices for Open X-haul transport infrastructure                                                                                                                                                                                   |
| WG10 - OAM for O-RAN                             | Develops OAM specifications for O1 interface and creates OAM architecture consistent with the overall OAM architecture specified by WG1.                                                                                                                           |
| WG11 - Security Work Group                       | Specification of requirements for O-RAN defined interfaces and network functions, security protocols, and study of security analysis of various components such as O-Cloud, SMO, Near-RT, and Non-RT RIC                                                           |

In addition to the above work groups, there are 4 focus groups.

- SDFG (Standard Development Focus Group): It works on the standardization strategies and main interface to other Standard Development Organizations (SDOs).
- TIFG (Test and Integration Focus Group): It defines O-RAN's approach for testing and integration, including coordination of test specifications across various WGs, planning and coordination of the O-RAN Alliance Plugfests and sets guidelines for the 3rd party Open Testing and Integration Centers (OTIC).
- OSFG (Open-Source Focus Group): This is a dormant group as opensource related activities happen in O-RAN Software Community.
- nGRG (next Generation Research Group): Research of open RAN principles in the context of 6G and future networks.



# O-RAN Architecture

The below diagram, from Reference [2], shows the O-RAN Architecture components and their corresponding 3GPP and O-RAN defined interfaces:

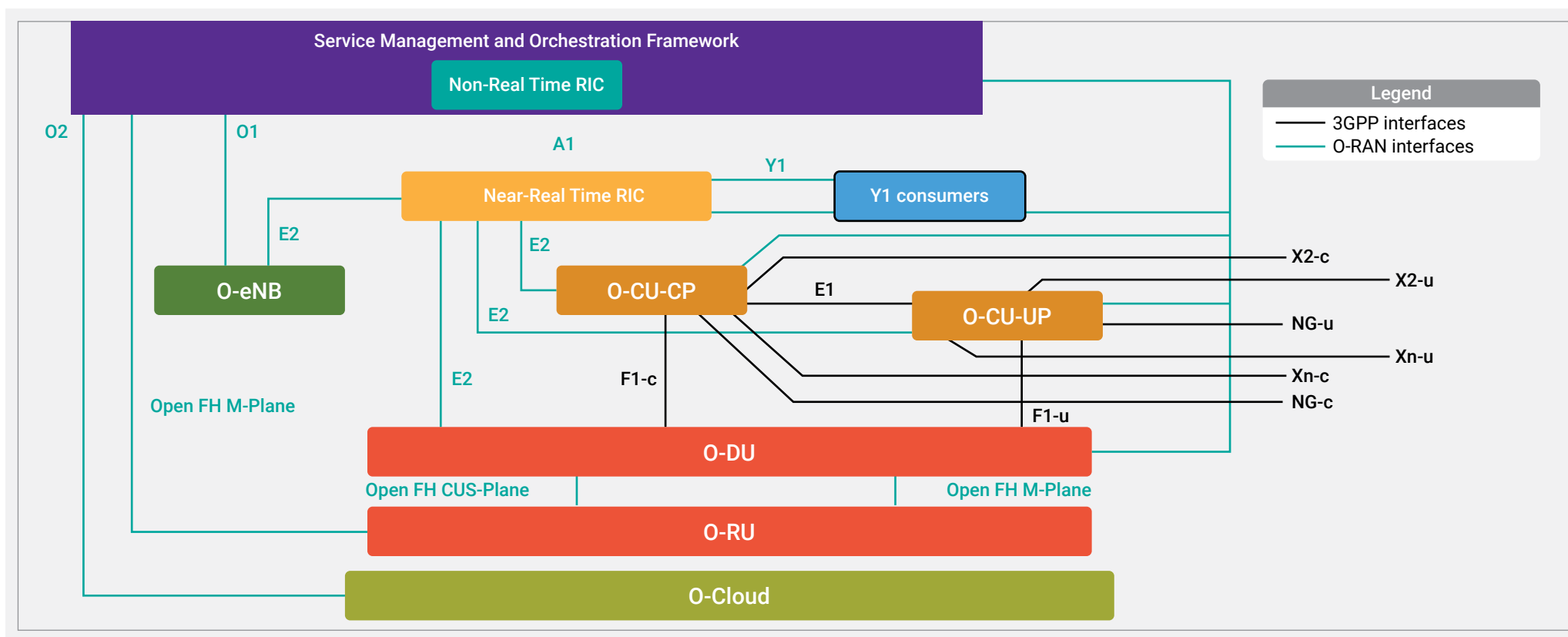


Figure 2: O-RAN Architecture



The components Near-Real Time RAN Intelligent Controller (Near-RT RIC), SMO, O-CU-CP, O-CU-UP, and O-DU, along with O-RU, define O-RAN Network Functions. These network functions can be implemented as Virtual Network Functions (VNFs) and/or Physical Network Functions (PNFs).

The description of various ORAN functions is given below:

- **Service Management and Orchestration (SMO):** It provides the following capabilities:
  - » Fault Configuration Accounting Performance Security (FCAPS) interface to O-RAN Network Functions
  - » Non-Real Time (Non-RT) RIC for RAN optimization
  - » O-Cloud Management, Orchestration Framework, and Workflow Management
- **Non-Real Time (Non-RT) RIC:** It is a logical function inside SMO to support intelligent RAN optimization via policy guidelines, Machine Learning (ML) model management, and sending information to the Near-RT RIC. It supports a latency of more than 1 second and is based on a microservices architecture where microservices (implementation of a single application as many loosely coupled and independently deployable smaller services) called rApps are deployed on top of a non-RT RIC platform. It performs the following functions:
  - » Non-real-time intelligent radio resource management.
  - » Policy optimization in RAN is achieved by providing policy-based guidance, model management, and enrichment information to the near-RT RIC function so that the RAN can be optimized. g., radio resource prioritization for certain categories of users (fire fighters, emergency health workers) in a certain sector at a certain time.
  - » Guidance, parameters, policies, and AI/ML models to support the operation of near-RT RIC. AI-enabled policies and ML-based models generate messages in non-RT RIC and are conveyed to the near-RT RIC for execution. AI-based antenna electrical tilt via network KPIs for coverage enhancement and increased network capacity



- **Near-RT RIC:** It is a logical function to intelligently manage and orchestrate the RAN. It supports a latency of less than 1 second and is based on a microservices architecture where microservices (xApps) are deployed on top of the Near-RT RIC platform. It can be deployed as a VNF/CNF. It performs the following functions:
  - » Near-real-time Control and Optimization: Handover Management, monitoring of traffic and radio conditions in real time, RAN Slicing, QOS Control, Enhanced Radio Resource Management (RRM) with AI/ML, Per-UE controlled load balancing, Radio Database Management, Interference Detection and Mitigation.
  - » Collection and maintenance of historical data on traffic, navigation, radio, and handover data.
  - » Monitoring, suspension, override, or control node via non-RT RIC policies
- **O-CU-CP:** It is a logical node for the implementation of RRC and the control plane part of the PDCP protocol.
- **O-CU-UP:** It is a logical node for the implementation of SDAP and the user-plane part of the PDCP protocol.
- **O-DU:** It is a logical node for the implementation of the RLC/MAC/higher PHY layers of the protocol stacks.
- **O-RU:** Depending on the architecture split between O-DU and O-RU implements either some, all, or none of the functionalities of the Layer 1 protocol stack along with the Radio Frequencies (RF part). Ex., O-RAN WG4 supports a Lower Layer Split (LLS), 7.2x, which requires part of the PHY (called L-PHY) to be implemented in O-RU and the rest of the PHY (called H-PHY) to be implemented in O-DU.
- **O-CLOUD:** O-Cloud is a cloud computing platform that consists of physical infrastructure for ORAN logical functions (Near-RT RIC, O-CU-CP, O-CU-UP, and O-DU) and supports software components (OS, Virtual Machine, Container), management, and orchestration functions.
- **O-eNB:** The open E NodeB for 4G RAN. O-eNB supports O-DU and O-RU functions with an open front-haul interface between them.



# ORAN Functional Splits

3GPP has studied various possible functional splits between the DU and CU in the context of new radio access technology development. Figure 3 below (from Reference [1], section 11.1.1) shows the functional split. There are 8 possible options depicted for the functional split between the CU and DU.

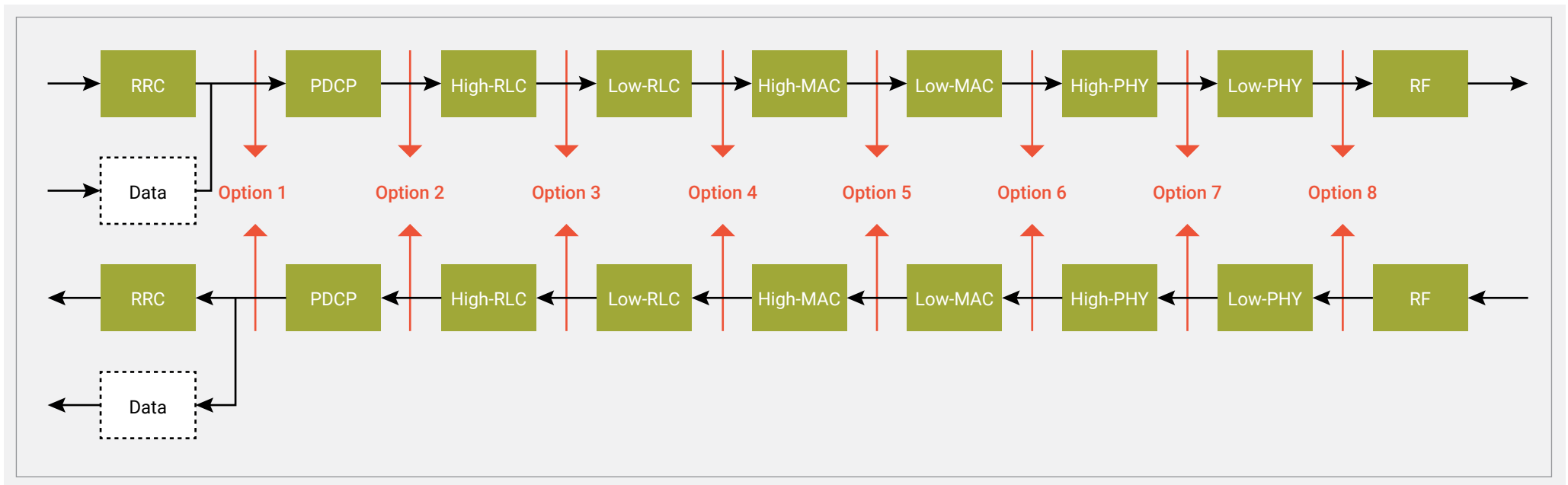


Figure 3: Functional Splits between the CU and DU



Option 1 to Option 5 are called “Higher Layer Split (HLS)”; Option 6 to Option 8 are called “Lower Layer Split (LLS)”. The decision to use a particular option depends on factors such as latency, bandwidth, and feature requirements. 3GPP has defined ‘option 2’ (HLS) between CU and DU. O-RAN WG4 has defined option 7.2x for the LLS, while O-RAN WG7 considers other LLS as an open front-haul interface for split base station architecture. As per WG7 Deployment Scenarios, a fronthaul interface that meets one of the following criteria can be defined as an open fronthaul interface [9]:

- O-RAN WG4 released interfaces OR
- O-RAN approved publicly (e.g., small cell forum, etc.) available interfaces OR
- Fronthaul interfaces are made available and published as part of an O-RAN-approved WG7 reference design.

To ensure interoperability between O-DU and O-RU supporting different options, a fronthaul Gateway (FHGW<sub>x→y</sub>) function for translating from split option x of O-DU<sub>x</sub> to split option y of O-RU<sub>y</sub> is needed. The following split option with a fronthaul translation is currently noted in WG7 for the deployment scenarios [9]:

- O-DU<sub>7.2x</sub> ↔ FHGW<sub>7.2x→8</sub> ↔ O-RU<sub>8</sub>

As a summary, following split options are recognized by O-RAN WG7 as per the identified deployment scenarios [9]:

| CU       | DU          |                        | RU       |
|----------|-------------|------------------------|----------|
| Option 2 | Option 6    |                        |          |
|          | Option 7.2x |                        |          |
|          | Option 8    |                        |          |
|          | Option 7.2x | FHGW <sub>7.2x→8</sub> | Option 8 |

Table 1: Functional splits

Note that Table 1 needs to be interpreted in the context of the deployed base station architecture (ex, in an integrated base station architecture, DU & RU are integrated into one box) [9]. There are variations within split option 7.2x depending on where the precoding option is located. If the precoding is not done in O-RU, then they are called ‘Category A’ (lower complexity) O-RUs, and if the precoding is done in O-RU, then they are called ‘Category B’ O-RUs. A high-level RAN functional diagram based on the above functional splits for HLS and LLS (option 7.2x) is shown below:



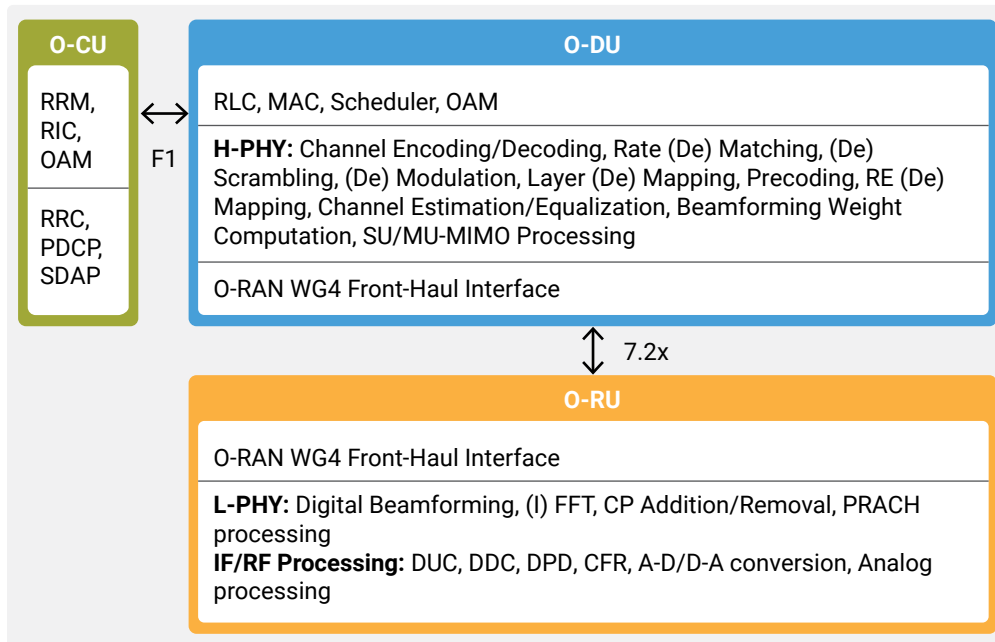


Figure 4: Open RAN Functional diagram based on LLS option 7.2x

### O-RAN Front Haul Interface

- The O-DU to O-RU interface, as defined by O-RAN WG4 (also known as the front haul interface [3]), is based on the eCPRI interface [4]. It is a packet-based interface and can be framed within Ethernet. The data flow between the O-DU and O-RU is categorized into the following planes of operation:
- C-Plane (Control Plane):

- » Scheduling and beam-forming commands: scheduling information, FFT size, CP length, subcarrier spacing, PRACH scheduling, DL and UL beam-forming commands, and scheduling.
- U-Plane (User Plane):
  - » DL I/Q Data: DL frequency domain user (PDSCH) and control channel data (e.g., PDCCH)
  - » UL I/Q Data: UL frequency domain user (PUSCH) and control channel data (e.g., PUCCH)
  - » PRACH I/Q Data
- S-Plane (Synchronization Plane): The Synchronization Plane is responsible for the timing and sync aspects between the O-DU and O-RU. Using S-Plane, O-RAN fronthaul specifications support protocols such as PTP and SyncE to achieve high-accuracy synchronization on the O-RU side by synchronizing with the high-performance clock available on the O-DU side.
- M-plane (management plane): Management plane messages are used to manage the radio unit. M-Plane provides a variety of O-RU management functions to set parameters on the O-RU side as required by the C/U-Plane and S-Plane, e.g., manage O-RU software, perform fault management, etc. The O-RAN fronthaul specification for M-Plane [5] provides various parameters as data models for FCAPS functions. These data models eliminate dependence on each O-RU vendor's implementation and make a real multi-vendor Open RAN possible.



# ORAN Benefits

## Lower TCO

- RAN is at least 60% of the total cost of ownership (TCO) for a network operator. The TCO includes CAPEX (deployment cost, which is 30%) and OPEX (operating cost, which is about 70%). As per estimates, OPEX savings can be in the range of 40 to 53% while CAPEX savings can be in the range of 30 to 50%. [7]

## OPEX

The OPEX costs can be reduced in the following ways:

- Zero-touch provisioning: no manual tasks to configure the cell sites
- AI/ML: Using a data analytics framework for optimizations (e.g., load balancing, handoff management).

## CAPEX

- The Capex costs can be reduced by a more competitive supply chain (due to multi-vendor interoperability).
- Minimizing the danger of vendor lock-ins: This benefits MNOs as it avoids vendor lock-in and enables interoperability.

# ORAN Deployments and Opportunities

The current ORAN deployments are from greenfield operators like 1&1 in Germany, Dish in the United States, and Rakuten Mobile in Japan. Telefonica is working with NEC on pre-commercial trials in Spain, Germany, the UK, and Brazil. Vodafone has launched the first commercial ORAN 4G site in a rural area at the Royal Welsh Showground in Powys, Wales, in the UK, with Mavenir, Dell, Kontron, and Lime Microsystems. It has also scheduled 2 pilot projects for 5G ORAN in rural Bavaria, Germany [6].

The ORAN market registered revenues of \$180 million in 2020. According to Stefan Pongratz, vice president of research firm Dell'Oro Group, operators globally spend around \$300 billion annually on capital outlays, a third of which goes towards network equipment. There's approximately an even split between RAN and mobile core network spend and other equipment in the \$100 billion [6]. According to a research report by Kenneth Research, the ORAN market opportunity is expected to reach a revenue of \$22.5 billion by 2028 with a CAGR of 85% during 2020-2028 [8].

Apart from existing vendors, there is an opportunity for ORAN in private networks, which are all essential greenfield deployments.



# ORAN Challenges

The most important challenge for ORAN is integration since it is a multi-vendor system. Integration here refers not just to connecting elements of ORAN and making sure that functionality works. It also includes software and hardware release management, version maintenance, KPI assurance, and observability in deployed networks and security. Since ORAN is an evolving system and not a mature system, integration cycles will take time, and this will impact the operator's plan for deployment. Apart from the need to have interoperability between O-RAN elements from different vendors (e.g., O-CU and O-DU from different vendors)—also called horizontal integration—integration also involves ensuring software components perform sufficiently well on selected platforms (e.g., DU software on a COTS)—also called vertical integration. The front-haul interface is the most prominent and challenging interface during the integration. It is an interface consisting of multiple logical planes (C, U, S, and M planes) and will be a challenge while integrating with new radio vendors to achieve the same KPI metrics. The RIC, along with xapps and rApps, are the most complicated parts of the ORAN systems. Based on AI/ML model execution, they send instructions back to RAN functions for performance optimization. To ensure third-party xapps and rApps work as expected and co-exist, a lot of RIC conformance testing needs to be done and testing these apps in the needed environment (simulation or a real network) is a challenge.

The other challenge is that most of the ORAN deployments are from greenfield operators. There are not many large-scale greenfield networks to be built. This in turn limits the market possibilities.

The current brownfield operators are unlikely to switch to ORAN systems unless there is a significant price advantage over existing systems and the performance and power consumption of ORAN systems are compatible with those of existing systems. As of 2022, the cost savings and the equivalent performance that are expected are still to be established.



# Conclusion

This whitepaper attempts to outline ORAN systems, their architecture, benefits, and challenges faced while deploying ORAN systems. The major positives noted are operators' ability to choose from diverse vendor ecosystem systems, TCO savings through network automation, and the opportunity for innovation. However, the challenges remain in terms of integration and the limited greenfield operators as of now. Time will tell whether ORAN causes a disruption in the market.

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9. [ORAN WG7, Deployment Scenarios and Base Station Classes, v04.00](#)

# Glossary

|       |                                        |
|-------|----------------------------------------|
| AI    | Artificial Intelligence                |
| CAPEX | Capital Expenditure                    |
| DL    | Downlink                               |
| eCPRI | Enhanced Common Public Radio Interface |
| I/Q   | In-phase/Quadrature                    |
| MAC   | Medium Access Control                  |
| ML    | Machine Learning                       |
| MNO   | Mobile Network Operator                |
| OPEX  | Operational Expenditure                |
| ORAN  | Open Radio Access Network              |
| PDCCP | Packet Data Convergence Protocol       |
| PTP   | Precision Time Protocol                |
| RF    | Radio Frequency                        |
| RLC   | Radio Link Control                     |
| RRC   | Radio Resource Control                 |
| RRM   | Radio Resource Management              |
| SDAP  | Service Data Adaptation Protocol       |
| TCO   | Total Cost of Ownership                |



## About the Authors

### **Amit Palkar**

Amit Palkar started his career at Sasken (then Silicon Automation Systems) in 1996. Beginning with the area of speech coding, he went on to play an important role in the development of Sasken's ADSL modem technology. He played a key role in designing the USB interface-based host ADSL modem. Some of the work in the wireless domain includes the architecture and design of the enodeB physical layer for O-RAN front haul compatibility with multiple radio vendors and modifications of the terrestrial enodeB stack for Non-Terrestrial Networks (NTN).

### **Venkata Murali Krishna Adapa**

Venkata Murali Krishna Adapa is an Architect with Sasken Technologies Ltd. with over 22 years of experience in architecting, designing, and implementing Radio Access Network (RAN) and modem systems based on multiple generations of 3GPP technologies. His recent contributions include architecture and design of the eNB physical layer for O-RAN FH compliance ensuring multi-vendor inter-operability, and the adaptation of the eNB protocol for Non-Terrestrial Networks (NTN) scenarios.

## About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to-market, chip-to-cognition R&D services to global leaders in the Semiconductor, Automotive, Industrials, Consumer Electronics, Enterprise Devices, SatCom, Telecom and Transportation industries. For over 30 years and Communications with multiple patents, Sasken has transformed the businesses of 100+ Fortune 500 companies, powering more than a billion devices through its services and IP.





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